

APPENDIX
OMITTED PROOFS

Proof of Theorem 1. Fix a constant $\psi \geq 1$. We give a gap-preserving reduction from INDEPENDENT SET on 3-regular graphs. Let $(H = (U, F), \ell)$ be an instance, where the question is whether H contains an independent set of size at least ℓ . Write $n_H := |U|$.

Choice of the gap parameter. Choose an even integer Q such that

$$Q > \psi\ell \quad \text{and} \quad Q \geq 8.$$

Since ψ is a fixed constant, Q is polynomial in the input size.

Katz scores. We use the Katz centrality vector

$$s_G = (I - \alpha A(G)^\top)^{-1} \mathbf{1},$$

with $\alpha = \frac{1}{2}$. Since the graph constructed below is a DAG of depth at most 3, all Katz scores can be computed by summing walks of length at most 3.

Construction. We construct a directed, unweighted graph G with two groups *red* and *blue*.

For each vertex $u_i \in U$, create:

- a selector tail t_i , colored *blue*;
- a candidate node r_i , colored *red*.

We make each selector tail have score exactly $Q + 1$ by adding $2Q$ private leaf nodes

$$y_{i,1}, \dots, y_{i,2Q},$$

colored *blue*, with fixed edges

$$y_{i,a} \rightarrow t_i \quad \text{for all } a \in [2Q].$$

Thus

$$s(t_i) = 1 + \alpha(2Q) = Q + 1.$$

For each edge $e = \{u_i, u_j\} \in F$, create ℓ conflict clones

$$q_{e,1}, \dots, q_{e,\ell},$$

colored *blue*, and add fixed edges

$$r_i \rightarrow q_{e,h}, \quad r_j \rightarrow q_{e,h}, \quad \text{for every } h \in [\ell].$$

Next create ℓ blue buffer nodes

$$w_1, \dots, w_\ell.$$

Each buffer node w_a is constructed to have Katz score

$$s(w_a) = \frac{Q}{2} + \frac{3}{4}.$$

To achieve this, add $Q - 2$ private leaves

$$d_{a,1}, \dots, d_{a,Q-2}$$

with fixed edges $d_{a,b} \rightarrow w_a$, and add one intermediate node p_a with one private leaf $z_a \rightarrow p_a$ and one fixed edge $p_a \rightarrow w_a$. Then $s(p_a) = 1 + \alpha = \frac{3}{2}$, and therefore

$$s(w_a) = 1 + \alpha(Q - 2) + \alpha \cdot \frac{3}{2} = 1 + \frac{Q - 2}{2} + \frac{3}{4} = \frac{Q}{2} + \frac{3}{4}.$$

Finally, create ℓ fallback red nodes

$$f_1, \dots, f_\ell,$$

colored *red*. For each fallback node f_a , create Q private blue leaves

$$g_{a,1}, \dots, g_{a,Q}.$$

The edges $g_{a,b} \rightarrow f_a$ will be admissible edits rather than fixed edges.

Admissible edits. The admissible action set $\mathcal{A}$ consists of two types of edges:

- core selector edges

$$t_i \rightarrow r_i \quad \text{for every } u_i \in U;$$

- fallback edges

$$g_{a,b} \rightarrow f_a \quad \text{for every } a \in [\ell], b \in [Q].$$

There are no other admissible edits.

Top- k parameters. Set

$$k := n_H + \ell, \quad \pi := \frac{\ell}{k}, \quad \varepsilon := \frac{1}{4k^2}.$$

Because the number of red nodes in the top- k is an integer, any nonzero deviation from the target red count ℓ is at least $1/k$ in proportion. Hence

$$\text{Unfair}(T_k(G \oplus \Delta)) < \varepsilon$$

implies

$$|T_k(G \oplus \Delta) \cap V_{\text{red}}| = \ell.$$

Thus feasibility for the dual instance is equivalent to achieving exactly ℓ red nodes in the top- k set.

Score levels. Let G' be any graph obtained by adding admissible edits. For a core candidate r_i , there is at most one admissible incoming selector edge $t_i \rightarrow r_i$. Thus:

$$s(r_i) = \begin{cases} 1, & \text{if } t_i \rightarrow r_i \text{ is not added,} \\ 1 + \alpha s(t_i) = 1 + \frac{Q+1}{2} = \frac{Q}{2} + \frac{3}{2}, & \text{if } t_i \rightarrow r_i \text{ is added.} \end{cases}$$

We say that u_i is selected if the edge $t_i \rightarrow r_i$ is added.

Consider a conflict clone $q_{e,h}$ for $e = \{u_i, u_j\}$. If neither endpoint is selected, then

$$s(q_{e,h}) = 1 + \alpha(1 + 1) = 2.$$

If exactly one endpoint is selected, then

$$s(q_{e,h}) = 1 + \alpha\left(\frac{Q}{2} + \frac{3}{2} + 1\right) = \frac{Q}{4} + \frac{9}{4}.$$

If both endpoints are selected, then

$$s(q_{e,h}) = 1 + \alpha\left(\frac{Q}{2} + \frac{3}{2} + \frac{Q}{2} + \frac{3}{2}\right) = \frac{Q}{2} + \frac{5}{2}.$$

For a fallback node f_a , if exactly r of its Q private fallback edges have been added, then

$$s(f_a) = 1 + \frac{r}{2}.$$

In particular, if $r < Q$, then

$$s(f_a) \leq 1 + \frac{Q-1}{2} = \frac{Q}{2} + \frac{1}{2},$$

whereas if all Q private fallback edges are added, then

$$s(f_a) = 1 + \frac{Q}{2} = \frac{Q}{2} + 1.$$

For $Q \geq 8$, the relevant score levels satisfy the strict ordering

$$Q + 1 > \frac{Q}{2} + \frac{5}{2} > \frac{Q}{2} + \frac{3}{2} > \frac{Q}{2} + 1 > \frac{Q}{2} + \frac{3}{4} > \frac{Q}{2} + \frac{1}{2} > \frac{Q}{4} + \frac{9}{4} > 2 > 1.$$

Therefore the n_H selector tails t_i are always the highest-scoring nodes in the graph and hence always belong to the top- k set. The remaining ℓ positions of the top- k set are determined by the nodes below the selector tails.

YES case. Suppose H contains an independent set $S \subseteq U$ with $|S| \geq \ell$. Choose any subset $S' \subseteq S$ of size exactly ℓ . Add the ℓ selector edges

$$\{t_i \rightarrow r_i : u_i \in S'\}.$$

Every selected red candidate r_i has score $\frac{Q}{2} + \frac{3}{2}$. Since S' is independent, every conflict clone has at most one selected endpoint and therefore has score at most

$$\frac{Q}{4} + \frac{9}{4} < \frac{Q}{2} + \frac{3}{4} < \frac{Q}{2} + \frac{3}{2}.$$

The buffer nodes have score $\frac{Q}{2} + \frac{3}{4}$, and all fallback nodes are unactivated and have score 1. Thus, after the n_H selector tails, the next ℓ highest-scoring nodes are precisely the ℓ selected red candidates. Consequently,

$$|T_k(G \oplus \Delta) \cap V_{\text{red}}| = \ell,$$

and hence

$$\text{Unfair}(T_k(G \oplus \Delta)) = 0 < \varepsilon.$$

Therefore

$$\text{OPT}_\varepsilon \leq \ell.$$

NO case. Suppose H contains no independent set of size ℓ . We show that no feasible edit set of size smaller than Q exists.

Let Δ be any edit set with $|\Delta| < Q$. Since each fallback node requires all Q of its private fallback edges to reach score $\frac{Q}{2} + 1$, no fallback node is activated by Δ . Indeed, every fallback node has score at most

$$\frac{Q}{2} + \frac{1}{2} < \frac{Q}{2} + \frac{3}{4},$$

which is below the blue buffer score.

Let

$$S_\Delta := \{u_i \in U : t_i \rightarrow r_i \in \Delta\}$$

be the set of core vertices selected by Δ .

If $|S_\Delta| < \ell$, then fewer than ℓ red core candidates have score above the buffer nodes, and no fallback red node has score above the buffers. Thus, among the ℓ top- k positions below the selector tails, fewer than ℓ are red. Hence the top- k set does not contain exactly ℓ red nodes.

Now suppose $|S_\Delta| \geq \ell$. Since H has no independent set of size ℓ , the set S_Δ cannot be independent. Therefore there exists an edge $e = \{u_i, u_j\} \in F$ with both endpoints in S_Δ . For this edge e , all ℓ conflict clones

$$q_{e,1}, \dots, q_{e,\ell}$$

have score

$$\frac{Q}{2} + \frac{5}{2},$$

which is strictly larger than the score $\frac{Q}{2} + \frac{3}{2}$ of every selected red candidate. After the n_H selector tails, the next ℓ highest-scoring nodes are these ℓ blue conflict clones. Thus the top- k set again does not contain exactly ℓ red nodes.

In both cases, Δ is infeasible. Since Δ was arbitrary with $|\Delta| < Q$, every feasible solution in the NO case has size at least Q :

$$\text{OPT}_\varepsilon \geq Q.$$

Feasibility of the NO instances. The constructed dual instance is always feasible. Indeed, by adding all Q private fallback edges into each of the ℓ fallback nodes, each fallback node obtains score

$$\frac{Q}{2} + 1 > \frac{Q}{2} + \frac{3}{4},$$

which is above the buffer score. If no core selector edges are added, then all conflict clones have score 2, and the ℓ activated fallback nodes occupy the ℓ positions below the n_H selector tails. Therefore the resulting top- k set contains exactly ℓ red nodes. Hence $\text{OPT}_\varepsilon < \infty$ for every constructed instance.

Gap and approximation contradiction. We have shown:

$$\text{YES instance:} \quad \text{OPT}_\varepsilon \leq \ell,$$

whereas

$$\text{NO instance:} \quad \text{OPT}_\varepsilon \geq Q.$$

By construction, $Q > \psi\ell$.

Now suppose there were a polynomial-time ψ -approximation algorithm for the dual problem. On a YES instance, it would return a feasible edit set of size at most

$$\psi \text{OPT}_\varepsilon \leq \psi\ell < Q.$$

On a NO instance, every feasible edit set has size at least Q , so the algorithm must return a solution of size at least Q . Thus, by checking whether the returned solution has size smaller than Q , we could decide whether H contains an independent set of size at least ℓ .

This would solve INDEPENDENT SET on 3-regular graphs in polynomial time, contradicting $P \neq NP$. Therefore no polynomial-time ψ -approximation exists unless $P = NP$. $\square$

Proof of Corollary 1. We assume rational input parameters with polynomial bit complexity; Katz scores can then be computed exactly by solving a rational linear system, and score comparisons can be performed in polynomial time. The problem is in NP: given an edit set Δ , we can check in polynomial time whether $|\Delta| \leq C$ and whether the edited graph satisfies the target fairness constraint. NP-hardness follows from the gap construction in Theorem 1. In particular, distinguishing whether $\text{OPT}_\varepsilon \leq \ell$ or $\text{OPT}_\varepsilon \geq Q$ is NP-hard, and therefore so is the threshold decision problem. $\square$

Proof of Proposition 1. We have $K_G = \sum_{k=0}^{\infty} \alpha^k A_G^k$ and $K_{G'} = \sum_{k=0}^{\infty} \alpha^k A_{G'}^k$. Adding the edge $u \rightarrow v$ yields $A_{G'} = A_G + E$, where E is the matrix with a single 1 in position (u, v) and zeros elsewhere. Hence $A_{G'} \geq A_G$ entrywise. For nonnegative matrices, entrywise inequality implies $\rho(A_{G'}) \geq \rho(A_G)$, so $\alpha < 1/\rho(A_{G'}) \leq 1/\rho(A_G)$ and both Neumann series converge. We prove $A_{G'}^k \geq A_G^k$ entrywise for all $k \geq 0$ by induction. For $k = 0$, both are the identity matrix. Assuming $A_{G'}^k \geq A_G^k$ entrywise,

$$A_{G'}^{k+1} = A_{G'} \cdot A_{G'}^k \geq A_G \cdot A_{G'}^k \geq A_G \cdot A_G^k = A_G^{k+1},$$

where the first inequality uses $A_{G'} \geq A_G$ and the second uses the induction hypothesis. Multiplying by $\alpha^k \geq 0$ and summing over k yields $K_{G'} \geq K_G$ entrywise. $\square$

Proof of Lemma 1. Adding $u \rightarrow v$ gives $A_{G'} = A_G + \mathbf{e}_u \mathbf{e}_v^\top$. By Sherman–Morrison [?] with $M = I - \alpha A_G$, $\mathbf{x} = \alpha \mathbf{e}_u$, $\mathbf{y} = \mathbf{e}_v$:

$$\begin{aligned} K_{G'} &= (M - \mathbf{x} \mathbf{y}^\top)^{-1} = K_G + \frac{\alpha K_G \mathbf{e}_u \mathbf{e}_v^\top K_G}{1 - \alpha \mathbf{e}_v^\top K_G \mathbf{e}_u} \\ &= K_G + \frac{\alpha}{1 - \alpha K_G(v, u)} \mathbf{k}_{\cdot u} \mathbf{k}_{v \cdot}. \end{aligned}$$

The denominator is positive: since $\alpha < 1/\rho(A_{G'})$, the rank-one matrix $\mathbf{k}_{\cdot u} \mathbf{k}_{v \cdot}$ is nonnegative by Proposition 1, so $\alpha/(1 - \alpha K_G(v, u)) > 0$, giving $1 - \alpha K_G(v, u) > 0$.

Finally, we have $\Delta s(a \mid u \rightarrow v) = \mathbf{1}^\top (K_{G'} - K_G) \mathbf{e}_a = \frac{\alpha}{1 - \alpha K_G(v, u)} (\mathbf{1}^\top \mathbf{k}_{\cdot u}) (\mathbf{k}_{v \cdot} \mathbf{e}_a) = \frac{\alpha s_G(u) K_G(v, a)}{1 - \alpha K_G(v, u)}$. $\square$

Proof of Proposition 2. Apply Equation (3) to each term in Definition 2 and factor. $\lambda_G(u, v) > 0$ since $\alpha > 0$, $s_G(u) > 0$, and $1 - \alpha K_G(v, u) > 0$ under the Katz-validity assumption. $\square$